

1-C:

- A – External cost minus external benefit gives **net external cost**.
- B – We **cannot** subtract external cost from private cost.
- C – This is correct because **external-cost plus private cost** will give “**social cost**”
- D – This is incorrect because social cost minus social benefit will give **net social cost**.

2-C:

- A – Marginal cost pricing is when $\text{marginal cost} = \text{price (Average revenue)}$, which means that **allocative efficiency** is achieved.
- B – Pollution permits restricting the pollution of rivers by private firms is again a regulation, which will help to reduce pollution, that is an external cost. Therefore, it helps to correct “market failure”
- C – Requiring firms to pay a minimum wage is a law, imposed in labor markets, which is designed to improve the problem of inequalities, and so , it will not **correct “market failure”**, thus making this correct.
- D – Non-provision of public goods by the private sector , because of the “free-rider problem” results in market failure. And when the government intervenes by providing the complete output of public goods at zero price, the **market failure is rectified**.

3-A:

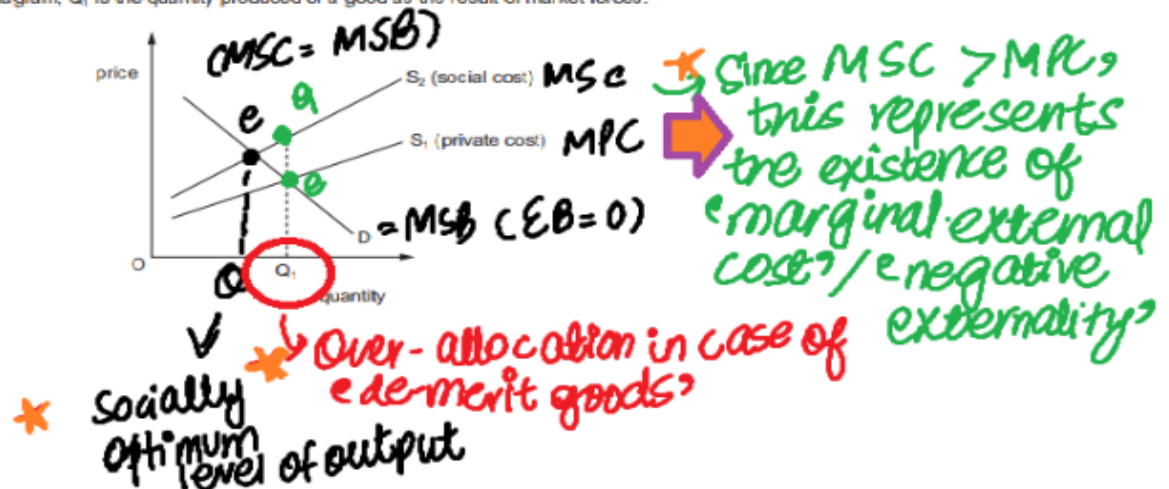
- The social cost is $(450(PC) + 75(EC))$ i.e. **“525”**
- The social benefit is **“550”**
- Since the social benefit is greater than the social cost, there is a net social benefit of “25”, which means that this project is socially worthwhile, and because of which, the “government” would be willing to build it.

Also,

- The private cost is “450”. And, the “private benefit is 500” $((550(SB) - 50(EB))$
- Since, the private benefit is greater than the private costs, the private sector , who is only interested in private costs and benefits, will also be willing to build it.

4-B:

In the diagram, Q_1 is the quantity produced of a good as the result of market forces.



A – Since, external benefits are zero, this represents the case of “demerit goods”, and subsidies are only provided to “merit goods”

B – Since, the particular good is a de-merit good, which is also over-allocated, there is an existence of marginal external costs or negative externalities, shown through “ae”, in figure#1. This is why, key B is correct.

C - Since, external benefits are zero, this represents the case of “demerit goods”, and positive externalities or external benefits are only present in case of “merit goods”.

D – If a specific tax was provided, then the supply curve would have shifted leftwards from $S_1(MPC)$ to $S_2(MPC+tax=MSC)$, and the output would have decreased from Q_1 to Q .

5-A:

Changes in budget lines:

- ➔ Upon changes in prices, the budget line tilts or pivots inwards or outwards.
- ➔ Upon changes in real-income, the budget line shifts leftwards or rightwards parallelly.

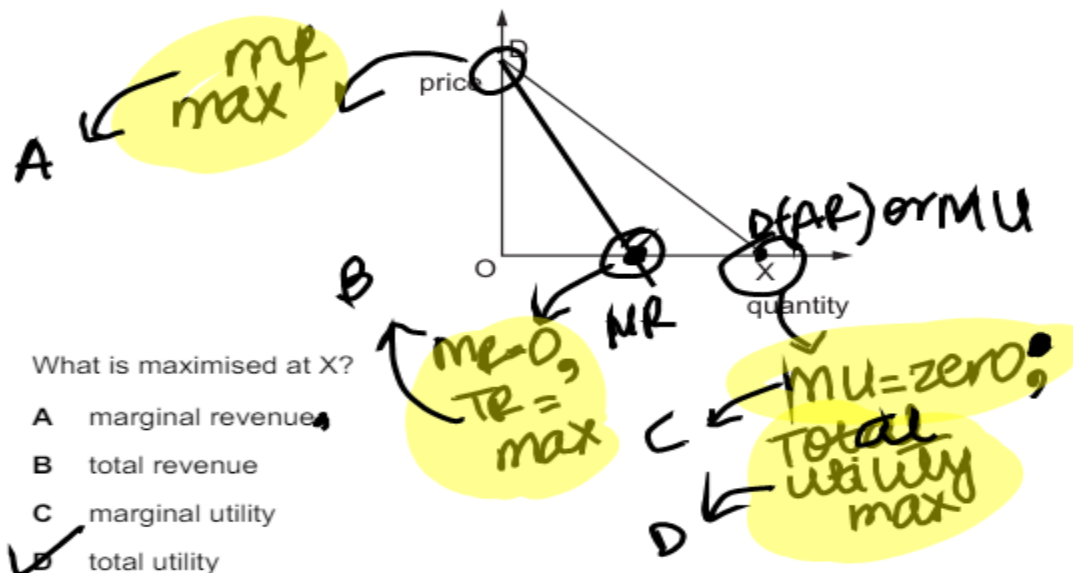
A – Individual preferences for various goods will affect the preference curve or the indifference curve, and not the budget line.

B – Changes in the level of income tax will bring about changes in the disposable income, thereby causing the budget line to shift.

C – Changes in the money prices of goods will cause the budget line to tilt.

D – Changes in wages earned by individuals will bring about changes in incomes, thereby causing the budget line to shift.

6-D:



The demand curve also represents the marginal utility curve (MU) when utility is measured in monetary terms.

At point X, since Marginal utility is equal to zero, this represents that Total utility will be maximized.

7-D:

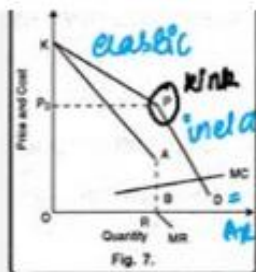
This is correct because a “**kinked demand curve**” represents the individual firm’s expectations about other firm’s responses to changes in its prices.

If for example, the firm raises its price, no other firm in the industry will follow, and therefore, this represents “**elastic demand**”(because, it is the behavior of a firm), and as a result, there is a contraction in quantity demanded by a relatively higher percentage, which results in a “fall in Total Revenue”.

Importantly, when a firm decreases its price, all other firms in the industry will follow, and therefore, this represents “**inelastic demand**”(because, it is the behavior of an industry), and as a result, there will be an extension in qty demanded by a relatively smaller percentage, which results in a “fall in Total Revenue”

Therefore, prices remain rigid or unchanged at the “kink”

D) It can be seen below, that the kinked demand curve, explains the concept of **price rigidity and interdependence**.



Price rigidity & Interdependence

✦ The kinked demand curve is a theoretical model to understand how firms might respond to changes in prices
The key features of the kinked demand curve are as follows:

✦ **Flat and Elastic Range:** Above the kink, the demand curve is relatively elastic. It suggests that if the firm raises its price slightly in an hope to increase total revenues, it will lead to a significant decrease in the quantity demanded by consumers and subsequently, a fall in total revenue, because other firms will not follow, which represents "elastic demand" (since it is a behaviour of one firm) As a result, the firm would lose a substantial portion of its market share.

✦ **Interdependence** refers to the mutual influence and dependence among firms in an oligopoly market. Unlike perfectly competitive markets, where firms are price takers and have no impact on market prices, in an oligopoly, each firm's actions and decisions can significantly affect its competitors and the overall market.
In the case of the **kinked demand curve**, firms in a non-collusive oligopoly closely observe and react to each other's pricing and output decisions. They are aware that any change in their price or production levels will trigger responses from their rivals. This interdependence means that firms must consider how their competitors might react to their actions.

Therefore, the prices remain "rigid" at the "kink".

This is because of the concept of "interdependence"

✦ **Steep and Inelastic Range:** Below the kink, the demand curve is relatively inelastic. It implies that if the firm lowers its price slightly, the change in the quantity demanded will be relatively small, or in other words, the qty demanded will extend by a relatively smaller percentage, because other firms in the industry will follow (representing "inelastic demand"), which results in a fall in total revenue. Competing firms are expected to follow the price reduction, resulting in losses faced by everyone.

This is why, key D, is correct.

- ➔ **Economies of scale** represent the cost advantages in form of a **fall in Long-Run Average costs**, as a result of a **firm growing large in size**. This is not represented through the "kinked demand curve model"
- ➔ **Allocative efficiency** occurs when a firm engages in marginal cost pricing, which means that **Marginal cost (MC) is equal to price (P)**. This is not present under "imperfect competition"
- ➔ This is why, other options are incorrect.

8-C:

Airline fuel is a variable cost, since it will be used based on the number of flights. Therefore, a fall in the prices of airline fuel will cause the "variable cost" to decline.

Importantly, passenger aircraft is a fixed cost. When an airline purchases or leases a passenger aircraft, it incurs a significant upfront cost or ongoing lease payments. This cost does not directly depend on the number of passengers or flights operated by the airline. Therefore, a rise in the price of "passenger aircraft" will increase the fixed cost.

9-C:

C - Low fixed costs make it more likely that an industry is a contestable market. With low fixed costs, new firms can enter the market without incurring significant sunk costs, making it easier to challenge incumbents. Since because of this barriers will reduce, the contestability of the market will increase.

- Option A: Advertising that establishes consumer loyalty creates a barrier to entry (**market barrier**) as new firms would need to invest heavily to attract customers away from established brands.

- Option B: Research and development sharing among firms may create a barrier to entry, as new firms may not have access to this shared knowledge and expertise.

- Option D: Market rivals aiming to reduce product differentiation may create a barrier to entry, as new firms may struggle to compete with existing large scale producers of standardized products. **Benefits of economies of scale gained by existing firms will create a cost barrier in this case.**

Since options A, B and D increase barriers, the contestability of the market will reduce.

10-A:

A – If a bakery(that operates in the secondary sector) merges with a wheat farm(that operates in the primary sector and provides the bakery with raw materials that it requires, helping it to ensure a security of supplies), this is an example of a “**back-ward vertical integration**”

B – A car manufacturer buying a car showroom is an example of a forward-vertical integration since a firm in the secondary sector is merging with a firm in the tertiary sector

C _ A vineyard requires a farm of grapes and if he buys an apple farm, this is an example of “diversification”

D _ If two rival supermarkets integrate, this is an example of horizontal integration

11-D:

By definition , for “super-normal profits” to be earned, the Average revenue must be greater than the Average cost.

In both the cases ,where a profit-maximizing level of output is produced,(when **MC=MR**), and where sales revenue maximization has been done (when **MR=0**), supernormal profits are being made due to Average revenue being greater than Average cost, which can be **identified through the vertical distance being AR and AC.**

This explains why ,key D , will be correct.

12-B:

A – Limit pricing is when lower prices are set to deter entry in an imperfect market, mainly contestable markets. Here, no such behavior is demonstrated, which is why, key A, becomes incorrect.

B – Here, **price discrimination** is being done , as higher prices are charged from those who purchase seats one day before the flight (because, this represents the urgency to travel, perhaps for an important business meeting, and as a result, the **demand for such customers may be inelastic, which means that they are willing to pay a much higher price**), as compared to the ones who have purchased it months before, because it reflects that they may be leisure travelers, who are much price-sensitive, which is representative of them having an “**elastic demand**”

C - Price leadership occurs when one dominant firm in an industry sets the price, and other firms follow suit. The scenario does not indicate any firm setting the price and others adjusting accordingly. This is mostly seen in case of a “**dominant oligopoly**”

D – When Average cost = Average revenue in case of a **perfect competition**, this represents a situation of “normal profits” . “**Here, prices are changing , and firms are doing it, which means that this does not apply to “perfect competition”**”

13-B:

A: An emphasis of “sales maximization” represents the interests of “managers” (who are in control), which is contrary to the objectives of the owners (shareholders), who aim for profit maximization. This represents a divergence of interests, or in other words, a divorce of ownership and control.

B – When management salaries are linked to the long-run growth of the company’s share price, this means that the interests are taken in favor of both: the managers, who now get a satisfactory level of pay, and the shareholders, whose wealth will get maximized.

There is no situation of divorce of ownership and control because now the objectives of both these parties: the principal (shareholder) and the agent (manager) are linked.

C - The presence of X-inefficiency in a firm can indicate that the firm is being run in the interests of management, this is because, production decisions are taken by managers (agent) rather than shareholders (principal). X-inefficiency refers to the situation where a firm is not operating at its optimal level of efficiency and is incurring unnecessary costs or using resources inefficiently. This is because of lack of effective competition in an industry, as a result of which, the average costs are higher than they would be if the market was more contestable. The management may not effectively motivate or incentivize employees to perform at their best, resulting in lower productivity levels. This can lead to inefficiencies in production processes, delays, quality issues, and higher labor costs.

D - The decision to purchase artwork, is again taken by managers, which means that the shareholders are not informed about it. This again creates divergence of interests.

14-A:

* Total cost of employing 10 cleaners =
 $10 \times 6 = 60$

* Marginal cost (Change in Total cost)
of 11th cleaner

$$MC_{11} = TC_{11} - TC_{10} \quad (10 \times 6)$$

$$11.5 = (11 \times x) - 60$$

$$x = 6.5$$

* Initially \$6
was paid

* Now \$6.5 is
paid

* Increase of 0.5 \$

15-B:

A: An increase in indirect taxes would not cause a rise in the productivity of labor. Indirect taxes, such as sales taxes or value-added taxes, are levied on goods and services and can increase the cost of production for firms. This could lead to reduced supply, higher prices as a result of shortages and subsequently a contraction in quantity demanded.

B: Improvements in capital, such as new technology, machinery, or equipment, can increase the productivity of labor by enabling workers to produce more output with the same amount of effort, leading to a higher productivity.

C: A rise in consumer surplus would not cause a rise in the productivity of labor. Consumer surplus refers to the difference between the maximum price consumers are willing to pay for a good or service and the market price they actually pay.

D: A rise in the elasticity of supply of labor would not cause a rise in the productivity of labor. The elasticity of supply of labor measures how responsive the quantity of labor supplied is to changes in the wage rate. An increase in elasticity means that labor supply responds more strongly to changes in wages, but this does not directly affect productivity.

16-B:

→ Economic rent is the extra income a worker receives – above the minimum level they need in order to work. It is the area above the supply curve and below the wage rate.

Initially, the economic rent was “RTX”

Now, after a rise in the demand, the economic rent rises to “RUZ”

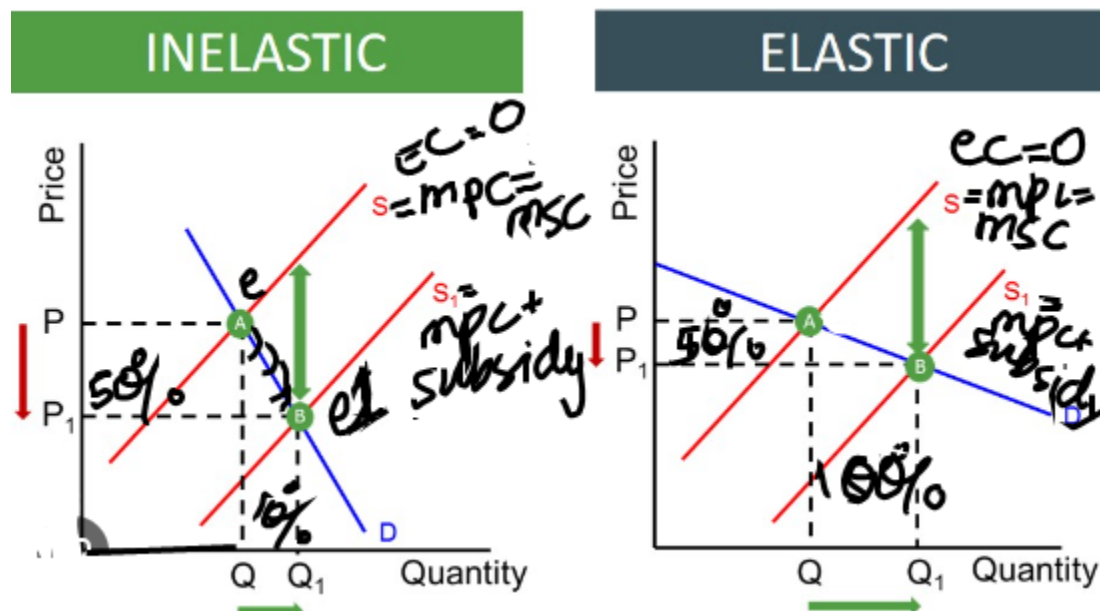
An increase in economic rent is shown through “TUZX”

This is why, key B, becomes correct.

17-D:

Governments do not provide subsidies to goods that generate negative externalities; instead they impose taxes on such products. This is why, options A and B will be incorrect.

Why option D and not C?



Subsidies represent a cash or a grant payment which when is provided to producers, brings about a reduction in costs of production, thereby leading to an increase in supply, or in other words, causes the supply curve to shift rightwards from S to S₁, as can be seen above. Because of a surplus and also producers attempts to pass benefits to consumers in form of lower prices, the equilibrium prices fall (by 50% in this case), as a result of which, there will be an extension in quantity demanded of merit goods by a relatively higher percentage (of 100%) when their demand is elastic(ped >1). This represents why subsidies will be the most beneficial when the ped is elastic.

18-D:

A: A constant rate sales tax would not help achieve a more equal distribution of income after tax. This tax is regressive, meaning it takes a larger share of income from lower-income individuals, who spend a larger proportion of their income on consumption. For example, if the sales tax rate is 10%, a person earning \$20,000 per year might spend \$10,000 on consumption, paying \$1,000 in sales tax (10% of \$10,000). Meanwhile, a person earning \$100,000 per year might spend \$30,000 on consumption, paying \$3,000 in sales tax (10% of \$30,000). The lower-income individual pays a larger share of their income in sales tax (5% vs 3%)

B : A constant rate tax per person would not help achieve a more equal distribution of income after tax. This tax is also regressive, as it takes the same amount from everyone, regardless of income level. For example, if the tax is \$1,000 per person, a person earning \$20,000 per year would pay 5% of their income in tax, while a person earning \$100,000 per year would pay only 1% of their income in tax.

C : An income tax with a constant marginal rate would not help achieve a more equal distribution of income after tax. **This tax takes the same percentage of income from everyone, regardless of income level.** For example, if the marginal rate is 20%, a person earning an additional income \$20,000 per year would pay \$4,000 in tax (20% of \$20,000), while a person earning additional \$100,000 per year would pay \$20,000 in tax (20% of \$100,000).

D :

$$\text{avg} \frac{0 + 2000}{20,000} \times 100 \Rightarrow 10\%$$

$$\text{avg} \frac{0 + 18000}{100,000} \times 100 \Rightarrow 18\%$$

← An income tax with a tax-free allowance and a constant marginal rate thereafter is the correct answer. This tax system allows a certain amount of income to be tax-free, and then applies a constant marginal rate to income above that threshold. For example, if the tax-free allowance is \$10,000 and the marginal rate is 20%, a person earning \$20,000 per year would pay no tax on the first \$10,000 and 20% on the remaining \$10,000, totaling \$2,000 in tax.

← A person earning \$100,000 per year would pay no tax on the first \$10,000 and 20% on the remaining \$90,000, totaling \$18,000 in tax. This system reduces the tax burden on lower-income individuals and increases the tax burden on higher-income individuals, helping to achieve a more equal distribution of income after tax.

* Since the average rate of tax is rising upon an increase in income, this represents a progressive tax system.

19-B:

Unemployment rate $\Rightarrow \frac{\text{Unemployed}}{\text{Labor force (Unemployed + employed)}} \times 100$

Let x represent unemployed

$$\hookrightarrow \frac{5}{5 + x} \times 100 \Rightarrow 10$$
$$\hookrightarrow 500 \Rightarrow 50 + 10x$$
$$x \Rightarrow \frac{500 - 50}{10} \Rightarrow 45$$

Total labor force $\Rightarrow 45 + 5 \Rightarrow 50$

20-B:

A and C— It is **seasonal unemployment** that arises as a result of a change in demand due to holiday periods or a temporary change in consumers' expenditure.

B — This is correct because a lack of sufficient information results in more time taken by workers to search a better job. This represents search unemployment which is a part of **frictional unemployment**.

D — A general decrease in demand for goods represents "**cyclical unemployment**"

21-A:

A - Nominal GDP per head only measures the total value of goods and services produced, without considering inflation or price changes. Real GDP, on the other hand, adjusts for inflation by using a base year's prices, allowing for a more accurate picture of economic growth.

B-This option is incorrect because real GDP does not ignore the effects of fluctuations in exchange rates on purchasing power.

C: This option is incorrect because real GDP does not account for changes in the size of the population.

D: This option is incorrect because real GDP does not measure GDP at factor cost. Instead, real GDP measures the value of goods and services produced at market prices.

22-D:

In a free market, open economy, the government does not intervene in economic activities, and there are no restrictions on trade. Autonomous investment (A) and household saving (B) are both economic activities that occur in a free market economy. Import spending (C) also occurs in an open economy, as individuals and businesses are free to import goods and services from other countries. Indirect taxation (D), on the other hand, is a government intervention in the economy, where taxes are imposed on goods and services. In a free market, open economy, there would be no indirect taxation, as the government does not interfere with economic activities.

23-C:

Unemployment benefits and pensions are transfer payments that are payments made without goods or services being received in exchange, and so, they do not result in any contribution to GDP.

The value of national income includes: wages (8000)+salaries(7000)+rent (3000)+ interest(2000). This sums up to \$20,000, as a result of this, option C becomes correct.

24-A:

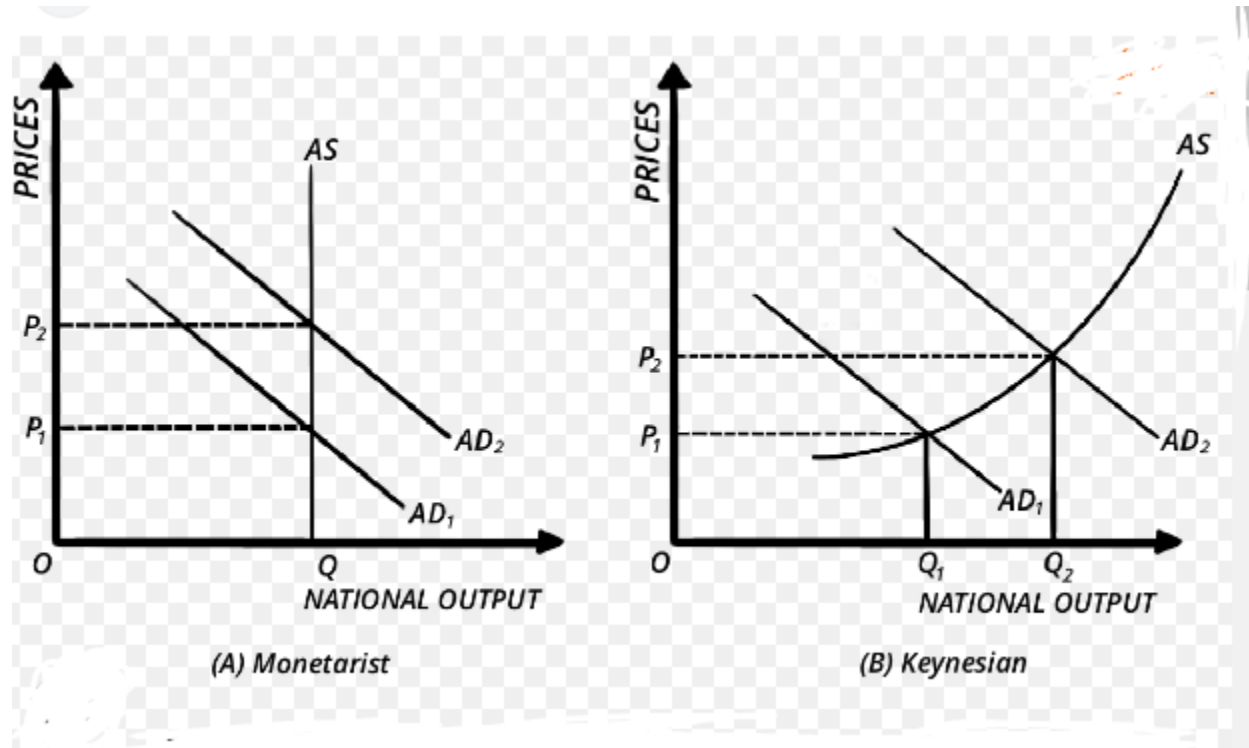
A) Keynesian economic analysis emphasizes the importance of aggregate demand in determining economic activity. A liquidity trap is a situation in which monetary policy (interest rates) becomes ineffective in stimulating economic activity, as people hoard cash and refuse to invest or consume. This is because the interest rates are too low and cannot fall further. This concept is a key part of Keynesian economics.

- B): This is a characteristic of classical economics, not Keynesian economics.

Keynesians argue that prices may not always adjust to clear markets.

- C): Keynesians believe in a large government expenditure multiplier, not a small one. **The multiplier effect suggests that government spending can have a significant impact on economic activity.**

- D) Keynesians believe in an upward-sloping short-run aggregate supply curve, not a vertical one. This can be seen in the diagram below.



25-D:

D) Increased government expenditure on education will lead to an increase in Aggregate demand, which will bring about a multiple times increase in Real national income that will lead to a rise in GNP.

A) This would reduce disposable income and lead to a decrease in consumption, which since will bring about a reduction in Aggregate demand, will decrease GNP.

B) This means that a greater proportion of additional income earned is being spent on imports and less is spent on domestically produced goods. This fall in consumption will bring about a reduction in Aggregate demand and subsequently GNP will decrease too.

- C) Increase in the value of a country's currency: An appreciation of the exchange rate makes export oriented goods relatively expensive and also makes imported goods relatively cheaper. This will lead to a Balance of Trade deficit in the long-run when the sum of combined elasticities of demand of imports and exports is greater than 1, which will bring about a reduction in AD and subsequently GNP will fall.

26-D:

A) The accelerator theory focuses on the change in last year's national income, not consumption.

B) While output is related to investment, the accelerator theory specifically looks at the change in last year's national income, not the level of output.

C) The theory looks at the change in last year's national income, not the change in investment.

D) According to the accelerator theory, this year's net investment is determined by the change in last year's national income. This means that if national income increased last year, businesses will invest more this year to meet the increased demand.

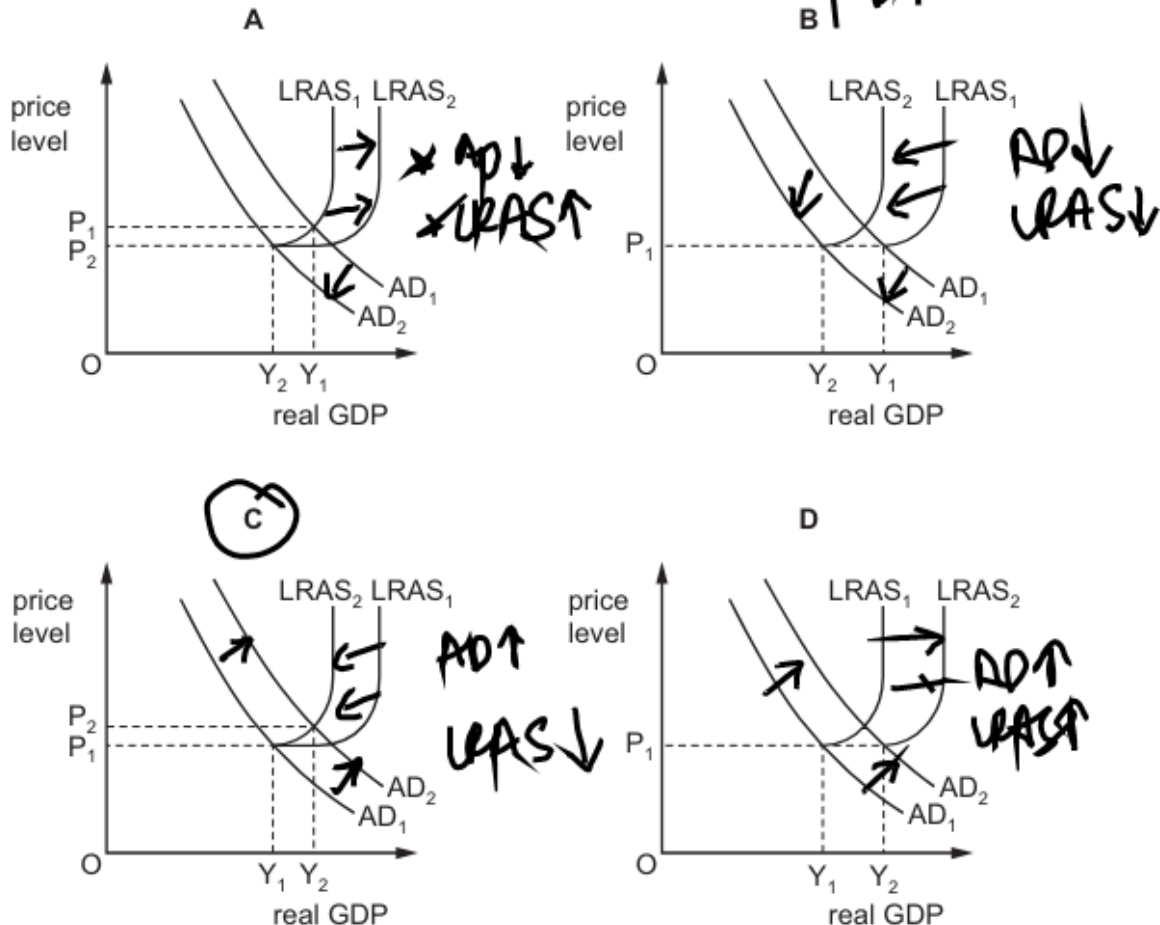
Multiplier $\rightarrow \frac{\Delta Y}{\Delta I} = \frac{500}{100}$ $\rightarrow 5$	Accelerator $\Rightarrow$ induced investment $\frac{\Delta I}{\Delta Y} \Rightarrow \frac{1500}{500}$ $\rightarrow 3$
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27-C:

Which diagram shows an economy experiencing an increase in actual output but a decline in potential output?

LRAS ↓

AD ↑



28-B:

A) Negative interest rates would not lead to an increase in government revenue.

Instead of interest rates, taxes are seen to directly influence tax revenues.

B) Negative interest rates are intended to stimulate borrowing and spending by incentivizing consumers when they borrow. This will lead to an increase in aggregate demand, and when an increase in Aggregate demand outpaces Aggregate supply, it becomes difficult for inelastic Aggregate supply to expand to meet rising Aggregate demand. This results in shortages, because of which, there is a rise in general price level or in other words, demand pull inflation. This is why, this becomes correct.

C) Negative interest rates are intended to encourage banks to lend more, not less.

D). Negative interest rates are actually intended to combat deflation, or falling prices arising from “too low of Aggregate demand” by stimulating economic activity and increasing borrowing and spending.

29-D:

A fall in CPI indicates deflation, which can lead to decreased aggregate demand and economic contraction.

A - This combination would be unlikely because an appreciation of the currency would reduce exports and increase imports, worsening the trade balance, and potentially leading to higher unemployment because of a fall in Aggregate demand. Additionally, increasing interest rates would reduce borrowing for investment purposes, further contracting the economy.

B - While increased government expenditure on education is an expansionary fiscal policy, pairing it with increased taxes would offset the stimulative effect. Taxes reduce disposable income, which would result in a decrease in consumption, thereby leading to a fall in aggregate demand and job creation. This combination would not effectively address high unemployment and deflation.

C - Depreciation makes exports cheaper and more competitive in the global market and it also increases demand for exports, potentially leading to increased export revenues and subsequently Aggregate demand, production and employment. But a rise in indirect taxation after having raised the prices of goods and services, will bring about a reduction in consumption and subsequently Aggregate demand.

D - This combination is the most likely because reduced income taxes increase disposable income, boosting consumption and aggregate demand. A cut in the rate of interest reduces borrowing costs and encourages investment, thereby boosting Aggregate demand. This combination aims to stimulate economic growth, combat deflation, and address high unemployment by increasing aggregate demand and job creation.

30-A:

A) When a country increases its interest rates, it makes investing in that country more attractive. This is because higher interest rates offer a higher return on investments. As a result, foreign investors are likely to be drawn to the country, leading to a **capital inflow**.

B) A depreciation of the currency is unlikely. Higher interest rates would attract foreign direct investment which will bring about an increase in demand for currency, typically lead to an appreciation of the currency.

C) An increase in consumption is unlikely, as higher interest rates would make borrowing more expensive, reducing consumption and aggregate demand.

D) An increase in investment is unlikely, as higher interest rates would make borrowing more expensive, reducing investment and aggregate demand.